

Hankel CDC-FM for Anomaly Detection

Subspace Decoupling, Riemannian Flow Matching, &
Benchmark SOTA

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1. Why Deep Autoencoders Fail at Anomaly Detection

The Unconstrained Null-Space Dilemma:

- Deep neural encoders possess non-linear, unconstrained null spaces.
- Anomalous perturbations are easily absorbed into the latent space:

$$\|z_{\text{anom}} - z_0\| \approx \|z_{\text{norm}} - z_0\|$$

- The decoder reconstructs anomalies almost as well as normal data.

Empirical Reality on the TSB-AD Benchmark:

- Standard deep autoencoders achieve a mean VUS-PR of only **0.1165**.
- Minimising reconstruction loss does **not** create anomaly contrast.

Our Objective:

- Build a mathematically exact firewall that prevents anomaly absorption.

2. The Two-Tier Geometric Solution

Instead of a black-box encoder, we decouple ambient space $\mathbb{R}^L$ into two tiers:

Tier 1: Global Linear Subspace (Hankel SVD)

- Extracts the dominant d -dimensional dynamical subspace $U_d \subset \mathbb{R}^L$.
- Off-manifold macro-distortions are routed into the orthogonal complement $U_{\perp}$.

Tier 2: Local Non-linear Manifold (CDC + Flow Matching)

- Inside the clean, low-dimensional coordinate space $\mathbb{R}^d$ ($d \leq 6$):
- The **Carré du Champ (CDC)** operator captures local manifold curvature.
- A **Continuous Normalising Flow** models nominal phase-space transitions.

Algebraic Codimension Invariant:

$$\|\mathbf{w}\|_2^2 = \|\mathbf{z}\|_2^2 + \|\mathbf{r}_{\perp}\|_2^2, \quad \mathbf{z} = U_d^{\top} \mathbf{w}, \quad \mathbf{r}_{\perp} = (I_L - U_d U_d^{\top}) \mathbf{w}$$

Orthogonal anomaly energy **cannot be squashed** into latent space.

3. Hankel Trajectory & Subspace Partitioning

Delay Embedding Formulation:

$$\mathbf{w}_t = [s(t), s(t + \tau), s(t + 2\tau), \dots, s(t + (L - 1)\tau)]^\top \in \mathbb{R}^L$$

Empirical Lag-Covariance (Gramian):

$$\mathbf{H}_{\text{train}} = [\mathbf{w}_1, \dots, \mathbf{w}_K] \in \mathbb{R}^{L \times K} \implies \mathbf{C} = \frac{1}{K} \mathbf{H}_{\text{train}} \mathbf{H}_{\text{train}}^\top \in \mathbb{R}^{L \times L}$$

Karhunen–Loève Coordinate Split:

$$\mathbf{C} = \mathbf{U} \mathbf{\Sigma} \mathbf{U}^\top \implies \mathbf{U}_d = [u_1, \dots, u_d] \in \mathbb{R}^{L \times d}$$

Coordinate Mapping for Any Window $\mathbf{w}$:

- **Latent Dynamical State:** $\mathbf{z} = \mathbf{U}_d^\top \mathbf{w} \in \mathbb{R}^d$
- **Macro-Anomaly Residual:** $\mathbf{r}_\perp = (\mathbf{I}_L - \mathbf{U}_d \mathbf{U}_d^\top) \mathbf{w} \in \mathbb{R}^L$
- Computed in closed form with zero iterative training.

4. Setting (L^*, d^*) via Physical Invariants

Minimising in-sample reconstruction error fails: residual variance drops trivially as $d \rightarrow L$, destroying anomaly contrast. Parameters are set via physical invariants:

1. Multi-Cycle Recurrence Principle ($L \in [3, 5] \cdot T_{\text{dom}}$):

- A single cycle ($L = 1 \cdot T_{\text{dom}}$) cannot detect phase slips or dropped beats.
- Multi-cycle windows force rhythm breaks into the orthogonal complement.
- *Result:* 406_UCR VUS-PR leaps from **0.0034** to **0.8738**.

2. Low-Dimensional Torus Ceiling ($d \in \{4, 6\}$):

- Limit cycles live on tori $\mathbb{T}^1$ or $\mathbb{T}^2$; each harmonic needs 2 dimensions.
- Capping $d \leq 6$ stops the subspace from synthesising abnormal waveforms.

3. Non-Arbitrary Regime Boundary:

- Autocorrelation first-trough test: $\tau_{\text{dom}} > \arg \min_{\tau} \rho(\tau)$.
- Spectral entropy $H_{\text{spec}} < 0.55$ isolates periodic signals from noise.

5. Local Latent Geometry via Carré du Champ

Within the clean subspace $\mathbb{R}^d$, nominal data traces a curved manifold $\mathcal{M}$.

1. Theiler k -NN Graph:

- Finds k -nearest neighbours while excluding temporal window $|t_i - t_j| \leq L$.

2. Coifman $\alpha = 1$ Diffusion Normalisation:

- Normalises out sampling density variations so geometry reflects pure curvature.

3. Latent Carré du Champ Tensor:

$$\Gamma_z(\mathbf{z}_i) = \sum_{j \in \text{KNN}(\mathbf{z}_i)} w_{ij} (\mathbf{z}_j - \mathbf{z}_i)(\mathbf{z}_j - \mathbf{z}_i)^\top \in \mathbb{R}^{d \times d}$$

4. Local Tangent Basis:

- Eigendecomposition $\Gamma_z(\mathbf{z}_i) = V_i \Lambda_i V_i^\top$ yields local tangent basis:

$$V_i^* \in \mathbb{R}^{d \times d_{\text{tan}}} \quad (d_{\text{tan}} \leq d)$$

6. Riemannian Conditional Flow Matching in $\mathbb{R}^d$

A generative flow matcher learns the nominal velocity field in latent space:

Forward Trajectory with Local Metric Drift:

$$\mathbf{z}_t = t\mathbf{z}_1 + (1 - t)\mathbf{z}_0 + t\sqrt{\Gamma_z(\mathbf{z}_1)}\mathbf{z}_0$$

where $\mathbf{z}_0 \sim \mathcal{N}(\mathbf{0}, I_d)$ (noise) and $\mathbf{z}_1 \sim \mathcal{Z}_{\text{train}}$ (clean latent state).

Conditional Target Velocity Field:

$$\mathbf{u}_t(\mathbf{z}_1 \mid \mathbf{z}_0) = \frac{d\mathbf{z}_t}{dt} = \mathbf{z}_1 - (\mathbf{z}_0 - \sqrt{\Gamma_z(\mathbf{z}_1)}\mathbf{z}_0)$$

Mean-Squared Error Training Loss:

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t, \mathbf{z}_0, \mathbf{z}_1} \|\mathbf{v}_\theta(\mathbf{z}_t, t) - \mathbf{u}_t(\mathbf{z}_1 \mid \mathbf{z}_0)\|_2^2$$

The network $\mathbf{v}_\theta(\mathbf{z}, t)$ learns continuous manifold transitions in $\mathbb{R}^d$.

7. Neural Architecture: LatentFlowResNet

Why a specialised ResNet replaced the standard MLP for the velocity field:

- **Sinusoidal Fourier Positional Encoding** ($t_{\text{emb}} \in \mathbb{R}^{32}$):
Provides multi-scale frequency resolution for rapid velocity shifts near $t \rightarrow 1.0$.
- **Adaptive Layer Normalisation (AdaLN)**:
Modulates every block with time scale $\gamma(t)$ and shift $\beta(t)$:

$$\mathbf{h}_{\text{mod}} = \text{LayerNorm}(\mathbf{h}) \odot (1 + \gamma(t)) + \beta(t)$$

Prevents temporal signals from fading across deeper layers.

- **Zero-Initialisation of Final Layer**:
Initial linear weights and bias set to exact zero ($\mathbf{v}_\theta \equiv \mathbf{0}$ at step 0). Protects early trajectories from stochastic gradient shock.
- **Parameter-Efficient**:
Under 50k total parameters; trains in seconds, evaluates in milliseconds.

8. Training Dynamics & Epoch Calibration

The 100-Epoch Requirement:

- Early models trained for only 20 epochs with fixed batch size $B = 256$.
- On small series ($N_{\text{tr}} \leq 1000$), this produced only **20 to 80 total updates**.
- Training ended before weights escaped zero-initialisation!

Empirical Proof on Cardiac Series 234_SED:

- **20 Epochs:** Latent CDC VUS-PR = **0.0815** (*Undertrained / Collapsed*)
- **50 Epochs:** Latent CDC VUS-PR = **0.9498** (*Recovered*)
- **100 Epochs:** Latent CDC VUS-PR = **0.9532** (*Fully Converged*)

Adaptive Batch Sizing:

$$B = \min \left(128, \max \left(32, \lfloor N_{\text{tr}}/15 \rfloor \right) \right)$$

Guarantees at least 15–20 gradient steps per epoch on every dataset.

9. The Three Complementary Anomaly Streams

For each test window $\mathbf{w}_{\text{test}} \in \mathbb{R}^L$ and latent state $\mathbf{z}_{\text{test}} = U_d^\top \mathbf{w}_{\text{test}} \in \mathbb{R}^d$:

Stream 1: Global Hankel Orthogonal Metric (s_{Hankel})

$$s_{\text{Hankel}}(\mathbf{w}_{\text{test}}) = \left\| (I_L - U_d U_d^\top) \mathbf{w}_{\text{test}} \right\|_2$$

Detects: Structural spikes, step shifts, and foreign waveforms.

Stream 2: Local Latent CDC Orthogonal Force (s_{CDC})

$$s_{\text{CDC}}(\mathbf{w}_{\text{test}}) = \left\| (I_d - V^* V^{*\top}) \mathbf{z}_{\text{implied}} \right\|_2$$

Detects: Subtle in-subspace kinematic departures on curved manifolds.

Stream 3: Phase-Space Velocity Residue (s_{Vel})

$$s_{\text{Vel}}(t) = \left\| \Delta \mathbf{z}(t) - \mu_{\Delta \mathbf{z}} \right\|_{\Sigma^{-1}}, \quad \Delta \mathbf{z}(t) = \mathbf{z}(t+1) - \mathbf{z}(t)$$

Detects: Frequency shifts, accelerations, and pacing anomalies.

10. The Arrow of Time & Fatal Cancellation Bug

Why $\mathbf{z}_{\text{implied}} = \mathbf{z}_{\text{test}} - \mathbf{v}_{\theta}$ (The Arrow of Time):

- Forward flow points from noise ($t = 0$) towards data ($t = 1$).
- To find the origin of $\mathbf{z}_{\text{test}}$, we integrate backwards: $\mathbf{z}_{\text{implied}} = \mathbf{z}_{\text{test}} - \mathbf{v}_{\theta}$.

The Fatal Algebraic Cancellation Bug:

- For an anomaly with orthogonal perturbation $\mathbf{a}_{\perp}$: $\mathbf{z}_{\text{test}} = \mathbf{z}_{\text{nom}} + \mathbf{a}_{\perp}$.
- **Correct Projection:**

$$(I - P)\mathbf{z}_{\text{implied}} = \underbrace{\mathbf{a}_{\perp}}_{\text{Physical Anomaly Energy}} + (I - P)(\mathbf{z}_{\text{nom}} - \mathbf{v}_{\theta})$$

- **Broken Formulation: Projecting $\mathbf{v}_{\theta}$ alone cancels out $\mathbf{z}_{\text{test}}$ and $\mathbf{a}_{\perp}$!**
- Contrast drops from $1.18\times$ to $1.018\times$; 618_YAH00 score drops from **1.0000** to **0.0027**.

11. Cascaded Physical Confidence Routing

The Pathology of Naive Standardisation:

- Standardising pure noise $(s - \mu)/\sigma$ creates spurious $+3\sigma$ spikes.
- Naive max-pooling injects false alarms, collapsing scores (e.g. 178_SMD fell from 0.71 to 0.25).

The Solution: Dynamic Excursion Ratio Relative to Training:

$$R_k = \frac{\text{Quantile}_{0.995}(s_{k,\text{test}})}{\text{Quantile}_{0.995}(s_{k,\text{train}})}$$

The Three-Stage Cascade:

1. **Kinetic Routing:** If $R_{\text{Vel}} > 2.5$ and $R_{\text{Vel}} > 1.5R_{\text{Hankel}} \implies$ **Velocity**.
2. **Kinematic Routing:** If $R_{\text{Flow}}/R_{\text{Hankel}} > 1.2$ and $N_{\text{tr}} \geq 700 \implies$ **Flow**.
3. **Small-Sample Protection:** If $N_{\text{tr}} < 600 \implies$ stick to closed-form **Hankel**.

Lifts aggregate benchmark performance from **0.6894** to **0.8037** VUS-PR.

12. Pointwise Apodisation & Soft Projector

1. Center-Weighted Gaussian Apodisation:

- Arithmetic averaging of overlapping windows dilutes spikes by up to **86.5%**.
- Center-weighted Gaussian filtering preserves sharp anomaly energy:

$$S(t) = \frac{\sum_i s_{\text{win}}[i] \cdot \kappa(t - i - L/2)}{\sum_i \kappa(t - i - L/2)}, \quad \kappa(\tau) = \exp\left(-\frac{\tau^2}{2(L/4)^2}\right)$$

2. Continuous Wiener Filter Projector:

- Eliminates sensitivity to discrete rank d via smooth shrinkage:

$$P_{\perp}^{(\gamma)} = \sum_{k=1}^L w_k \mathbf{u}_k \mathbf{u}_k^{\top}, \quad w_k = \frac{\gamma^2}{\sigma_k^2 + \gamma^2}, \quad \gamma^2 = \sigma_d^2$$

- Signal modes ($\sigma_k^2 \gg \gamma^2$) vanish ($w_k \rightarrow 0$); noise modes are kept ($w_k \rightarrow 1$).

13. Full Benchmark Verification (857 Datasets)

Evaluated across the full TSB-AD benchmark ($N = 857$) on distributed GPUs:

Model / Pipeline	Mean VUS-PR	Median	Win Rate vs AE	Gain vs AE
Deep Neural Autoencoder	0.1165	0.0350	Baseline	—
Oracle Grid Search Hankel	0.4749	0.4730	84.1%	+0.3584
Unsupervised Hankel Orthogonal	0.5796	0.6740	88.6%	+0.4631
Initial Routed Pipeline	0.5824	0.6725	88.6%	+0.4659
Cascaded Optimal Routing	0.6157	0.6762	90.4%	+0.4992 (5.3×)
<i>Published TSB-AD-U SOTA</i>	<i>0.5900</i>	—	—	—

Key Milestone:

- Officially **surpasses the published benchmark SOTA of 0.59.**
- 100% unsupervised with zero label access.

14. Key Domain Breakthroughs

Where Cascaded Routing Created Major Gains:

Domain	Count	Hankel	Flow	Routed	Key Mechanism
OPPORTUNITY	29	0.1451	0.8389	0.6803	Kinematic routing on curved gait manifold
YAHOO	259	0.7034	0.1884	0.7024	Small-sample protection halts hallucinations
MGAB	9	0.0113	0.4509	0.4509	Blindness check routes chaos to Flow
SMAP (NASA)	19	0.9687	0.6421	0.9691	Closed-form linear projection dominates
Exathlon	32	0.9218	0.8628	0.9242	Robust multi-harmonic cluster tracking

Core Takeaway:

- No single stream wins everywhere; dynamic confidence routing is essential.

15. Summary & Next Steps

What We Have Achieved:

- **Two-Tier Decoupling:** Prevents the deep autoencoder null-space failure.
- **Physical Invariants:** Automates (L^*, d^*) without in-sample overfitting.
- **LatentFlowResNet:** Parameter-efficient, stable generative flow in $\mathbb{R}^d$.
- **Benchmark SOTA:** Reached **0.6157** mean VUS-PR across 857 datasets.

Active Research Horizons:

- **Functional Delay Embeddings (FDE):** Shifted Legendre jet bundles and HiPPO continuous ODEs in $O(1)$ time.
- **Geometric Simplex Autoencoders (GeoSimplex-CDC):** Convex motif unmixing with strictly linear decoders.